Improving the EKF in some cases

- The EKF sometimes struggles to make good state estimates for systems having very nonlinear output equations.
- More refined linearization techniques can be used to reduce the linearization error and improve estimates.
- Some examples include the second-order EKF and Gaussian-sum filters, but we will not discuss those here.
- Instead, we will focus on a simple, common, and frequently helpful iterated-refinement approach.
- The result will be the algorithm known as the "iterated EKF" or "IEKF".

The EKF's output-equation linearization approach

- The EKF derivation approximated the measurement equation:

$$\begin{aligned} h(x_k, u_k, v_k) &\approx h(\hat{x}_k^-, u_k, \bar{v}_k) + \left.\frac{\partial h}{\partial x}\right|_{\hat{x}_k^-} (x_k - \hat{x}_k^-) + \left.\frac{\partial h}{\partial v}\right|_{\hat{x}_k^-} (v_k - \bar{v}_k) \\ &= h(\hat{x}_k^-, u_k, \bar{v}_k) + \widehat{C}_k (x_k - \hat{x}_k^-) + \widehat{D}_k (v_k - \bar{v}_k). \end{aligned}$$

- Based on this linearization, we then derived EKF steps 1c–2c to be:

$$\begin{aligned} \hat{z}_k &= h(\hat{x}_k^-, u_k, \bar{v}_k) \\ L_k &= \Sigma_{\tilde{x},k}^- \widehat{C}_k^T \left[\widehat{C}_k \Sigma_{\tilde{x},k}^- \widehat{C}_k^T + \widehat{D}_k \Sigma_{\tilde{v}} \widehat{D}_k^T \right]^{-1} \\ \hat{x}_k^+ &= \hat{x}_k^- + L_k (z_k - \hat{z}_k) \\ \Sigma_{\tilde{x},k}^+ &= \left(I - L_k \widehat{C}_k \right) \Sigma_{\tilde{x},k}^-. \end{aligned}$$

Reconsidering the EKF thought process

- The reason that we expanded $h(x_k, u_k, v_k)$ around $\hat{x}_k^-$ in the Taylor series was because that was our best guess of x_k before the measurement at time k is taken into account.
- But, after we compute $\hat{x}_k^+$, we have a better estimate of x_k than $\hat{x}_k^-$.
- So, after we compute $\hat{x}_k^+$, what if we re-linearize $h(\cdot)$ around $\hat{x}_k^+$ and recompute steps 1c-2c to see if we get an even better estimate of x_k?
- In fact, we could repeat this multiple times until we see some kind of convergence.
- This is the main idea of the IEKF.

Notation and initialization

- We introduce notation $\hat{x}_{k,i}^{+}$ to refer to the estimate of x_k after i re-linearizations have been performed.
  - We initialize $\hat{x}_{k,0}^{+} = \hat{x}_k^{-}$ to the prediction produced by the standard EKF.
- Similarly, $\Sigma_{\tilde{x},k,i}^{+}$ is the estimation-error covariance of $\hat{x}_{k,i}^{+}$ after i re-linearizations.
  - We initialize $\Sigma_{\tilde{x},k,0}^{+} = \Sigma_{\tilde{x},k}^{-}$, which is the prediction-error covariance produced by the standard EKF.
- We will denote the Kalman gain of the ith iteration as $L_{k,i}$, and the partial derivatives of the ith iteration as $\widehat{C}_{k,i}$ and $\widehat{D}_{k,i}$.

A first method to iterate the EKF

- As just mentioned, to begin the iteration process for every measurement update, we set $\hat{x}_{k,0}^{+} = \hat{x}_k^{-}$ and $\Sigma_{\tilde{x},k,0}^{+} = \Sigma_{\tilde{x},k}^{-}$.
- Then, for $i = 0, 1, \ldots, N$ we compute (in principle):

$$\begin{aligned}
\text{Step 1c}_i: \quad & \widehat{C}_{k,i} = \partial h/\partial x|_{\hat{x}_{k,i}^{+}} \quad \text{and} \quad \widehat{D}_{k,i} = \partial h/\partial v|_{\hat{x}_{k,i}^{+}} \\
& \hat{z}_k = h(\hat{x}_k^{-}, u_k, \bar{v}_k) \\
\text{Step 2a}_i: \quad & L_{k,i} = \Sigma_{\tilde{x},k}^{-}\widehat{C}_{k,i}^{T}\left[\widehat{C}_{k,i}\Sigma_{\tilde{x},k}^{-}\widehat{C}_{k,i}^{T} + \widehat{D}_{k,i}\Sigma_{\tilde{v}}\widehat{D}_{k,i}^{T}\right]^{-1} \\
\text{Step 2b}_i: \quad & \hat{x}_{k,i}^{+} = \hat{x}_k^{-} + L_{k,i}\,(z_k - \hat{z}_k) \\
\text{Step 2c}_i: \quad & \Sigma_{\tilde{x},k,i}^{+} = \left(I - L_{k,i}\widehat{C}_{k,i}\right)\Sigma_{\tilde{x},k}^{-}.
\end{aligned}$$

- Note that $\Sigma_{\tilde{x},k}^{-}$ is not changed in this modification to the EKF since it still describes the prediction; neither is $\hat{z}_k$ changed, but should it be?

A better way to iterate Step 1c

- The original EKF derivation used Assumption 1 to approximate:

$$\hat{z}_k = \mathbb{E}\left[h(x_k, u_k, v_k) \mid \mathbb{Z}_{k-1}\right] \approx h\left(\mathbb{E}[x_k \mid \mathbb{Z}_{k-1}], u_k, \mathbb{E}[v_k]\right) = h(\hat{x}_k^{-}, u_k, \bar{v}_k).$$

- An alternate derivation approach uses only Assumption 2 and not Assumption 1:

$$\begin{aligned}
\hat{z}_k &= \mathbb{E}\left[h(x_k, u_k, v_k) \mid \mathbb{Z}_{k-1}\right] \\
&\approx \mathbb{E}\left[h(\hat{x}_k^{-}, u_k, \bar{v}_k) + \left.\frac{\partial h}{\partial x}\right|_{\hat{x}_k^{-}}(x_k - \hat{x}_k^{-}) + \left.\frac{\partial h}{\partial v}\right|_{\hat{x}_k^{-}}(v_k - \bar{v}_k) \mid \mathbb{Z}_{k-1}\right] \\
&= h(\hat{x}_k^{-}, u_k, \bar{v}_k) + \widehat{C}_k\left(\mathbb{E}[x_k - \hat{x}_k^{-} \mid \mathbb{Z}_{k-1}]\right) \\
&= h(\hat{x}_k^{-}, u_k, \bar{v}_k) + \widehat{C}_k\left(\hat{x}_k^{-} - \hat{x}_k^{-}\right) = h(\hat{x}_k^{-}, u_k, \bar{v}_k).
\end{aligned}$$

- We will use this alternate approach to recompute $\hat{z}_{k,i}$ around $\hat{x}_{k,i}^{+}$ every iteration.
  - That is, we will compute $\hat{z}_{k,i} = h(\hat{x}_{k,i}^{+}, u_k, \bar{v}_k) + \widehat{C}_{k,i}\left(\hat{x}_k^{-} - \hat{x}_{k,i}^{+}\right)$.

Summary of IEKF steps

Step 1a: $\hat{x}_k^- = f(\hat{x}_{k-1}^+, u_{k-1}, \bar{w}_{k-1})$

Step 1b: $\Sigma_{\tilde{x},k}^- = \hat{A}_{k-1}\Sigma_{\tilde{x},k-1}^+ \hat{A}_{k-1}^T + \hat{B}_{k-1}\Sigma_{\widetilde{w}}\hat{B}_{k-1}^T$

Initialize iteration: $\hat{x}_{k,0}^+ = \hat{x}_k^-$ and $\Sigma_{\tilde{x},k,0}^+ = \Sigma_{\tilde{x},k}^-$

Iterate for $i = 0, 1, \ldots, N$

Step 1c$_i$: $\widehat{C}_{k,i} = \partial h/\partial x|_{\hat{x}_{k,i}^+}, \qquad \widehat{D}_{k,i} = \partial h/\partial v|_{\hat{x}_{k,i}^+}$

$\hat{z}_{k,i} = h(\hat{x}_{k,i}^+, u_k, \bar{v}_k) + \widehat{C}_{k,i}\,(\hat{x}_k^- - \hat{x}_{k,i}^+)$

Step 2a$_i$: $L_{k,i} = \Sigma_{\tilde{x},k}^- \widehat{C}_{k,i}^T \left[\widehat{C}_{k,i}\Sigma_{\tilde{x},k}^- \widehat{C}_{k,i}^T + \widehat{D}_{k,i}\Sigma_{\tilde{v}}\widehat{D}_{k,i}^T\right]^{-1}$

Step 2b$_i$: $\hat{x}_{k,i}^+ = \hat{x}_k^- + L_{k,i}\,(z_k - \hat{z}_{k,i})$

Step 2c$_i$: $\Sigma_{\tilde{x},k,i}^+ = \left(I - L_{k,i}\widehat{C}_{k,i}\right)\Sigma_{\tilde{x},k}^-.$

end of iteration “for” loop

Output: $\hat{x}_k^+ = \hat{x}_{k,N}^+$ and $\Sigma_{\tilde{x},k}^+ = \Sigma_{\tilde{x},k,N}^+$

Summary

- The EKF sometimes struggles to compute good state estimates if the output equation is “very nonlinear.”
- Examining the EKF algorithm, we notice that the output equation appears in Steps 1c–2c, which rely on $\hat{x}_k^-$ as the best “guess” of x_k found to date.
- But, by iterating Steps 1c–2c when we replace $\hat{x}_k^-$ with our best $\hat{x}_k^+$ found so far, we can improve estimates.
- We could iterate until some metric of convergence is satisfied, but often iterating only a few times (or even once) is helpful.
- In the next lesson, you will learn how to implement the IEKF in Octave, and see some results from a very difficult example.